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Research, teaching and performance evaluation in academia: the salience of quality

Simon Cadez*, Vlado Dimovski and Maja Zaman Groff

Department of Accounting and Auditing, Faculty of Economics, University of Ljubljana, Kardeljeva ploscad 17, 1000 Ljubljana, Slovenia

The workload of most academics involves two main activities: research and teaching. Despite the dual nature of the work, career advancement usually chiefly depends on research performance. Since academics are rational actors, warnings are beginning to emerge that current predominantly research-based performance evaluation systems may be detrimental to creativity and innovation in teaching. This paper investigates the substance of these warnings by revisiting the relationship between research performance and teaching quality. Using a large cross-disciplinary sample of academics within a research-oriented university, we find, consistent with prior evidence, that research productivity is not related to teaching quality, whereas research quality is positively related with teaching quality. These findings discount fears that research-based performance evaluation in academia may be detrimental to teaching quality.

Keywords: research productivity; research quality; teaching quality; performance evaluation; university

Introduction

In the academic world, few beliefs command more passionate allegiance than the view that teaching and research are harmonious and mutually beneficial activities (Galbraith and Merrill 2012; Ramsden and Moses 1992; Taylor 2007). However, ample and long-standing empirical evidence provides almost no support for a positive relationship between research and teaching performance. In their meta-analysis of 58 empirical studies, Hattie and Marsh (1996) find no such relationship. They conclude that the common belief that research and teaching are inextricably entwined is no more than a popular myth (Marsh and Hattie 2002). Recently, research attention has turned towards thinking how the relationship could, or should, work (Malcolm 2014).

While most universities emphasise their dual mission of teaching and research (Taylor 2007), formally stated goals are often not translated into their strategies and reward systems (Modell 2003). Performance evaluation in academia has traditionally been somewhat biased towards research indicators and it seems that this orientation is becoming even more profound (Bogt and Scapens 2012; Gendron 2008; Parker 2008). A growing number of reports document that universities striving for government funding, research grants and high rankings have adopted strategies including recruitment and reward systems that favour academics with top publications or the potential

*Corresponding author. Email: simon.cadez@ef.uni-lj.si

to secure such publications (Bogt and Scapens 2012; Douglas 2013; Gendron 2008; Hopwood 2008; Long et al. 2009; Parker 2012; Sangster 2011).

These developments hold significant implications for organisational (Besancenot, Faria, and Vranceanu 2009; Moll and Hoque 2011; Reidpath and Allotey 2010; Rodman, Biloslavo, and Bratoz 2013) and individual research strategies (Cadez 2013; Gendron 2008; Hopwood 2008). 'Publish or perish' is becoming the predominant strategy while 'teach or impeach' and 'publish and teach effectively' strategies (Hattie and Marsh 1996) are becoming marginalised. Besancenot, Faria, and Vranceanu (2009) and Bogt and Scapens (2012) warn that current performance evaluation advances may be detrimental to creativity and innovation in teaching. A research-based performance evaluation system provides an incentive to rational actors to economise on teaching efforts to improve their career prospects (Arnold 2008; Gendron 2008; Moya, Prior, and Rodriguez-Perez 2015).

This study responds to calls by Bogt and Scapens (2012) and Kallio and Kallio (2014) to investigate the nature and consequences of contemporary performance measurement within universities. More precisely, it investigates the substance of the warnings that research-based performance evaluation in academia is detrimental to quality teaching. As a relative novelty, following a call by Stack (2003), the study revisits the relationship between research and teaching quality by dividing these activities into productivity (quantity) and quality components (Long et al. 2009; Moed 2008). Hypotheses are tested using a large cross-disciplinary sample of academics in a university where performance evaluation is predominantly research based.

This study provides three important contributions. First, unlike most prior studies investigating the research/teaching relationship, it explicitly considers the role of a performance evaluation system in providing incentives and motivations for academics. Second, by splitting research performance into two dimensions, it challenges the conventional wisdom that research activity facilitates teaching effectiveness (Hattie and Marsh 1996) with a finding that only high-quality research enhances teaching quality. Third, warnings that research-based performance measurements may be detrimental to quality teaching are discounted based on the finding that high-quality research is actually congruent with quality teaching.

The paper is structured as follows. Following the introduction, contemporary performance evaluation trends in academia are outlined. Next, relevant literature is summarised and hypotheses are developed. This is followed by the research design and findings sections. The paper concludes with a discussion of the main findings and some implications.

The upsurge of research performance indicators in academia

The debate about the role of research and teaching in universities has a long history. In 1853, Newman declared that the university was a place of teaching universal knowledge and that its objective was the dissemination of knowledge rather than its advancement (Newman 1907, first published in 1858). If its objective were scientific discovery, Newman saw no reason why universities should have students. A century later, Robbins (1963) proposed a counter-view. He asserted that there is no borderline between teaching and research; they are complementary and overlapping activities.

It seems that Robbins' view has prevailed since. An overview of the current mission statements of the highest ranked universities reveals that both knowledge production (research) and knowledge dissemination (teaching/education/learning) are strongly

embedded as these universities' priorities. Some notable examples of such mission statements are:

The mission of the University of Cambridge is to contribute to society through the pursuit of education, learning, and research at the highest international levels of excellence.

New York University's mission is to be an international center of scholarship, teaching and research defined by a culture of academic excellence and innovation.

(Source: The webpages of the respective universities)

The general performance management literature advocates that to secure the effective implementation of strategy (Cadez and Guilding 2008) an organisation's performance measures should be aligned with its strategic objectives (Chenhall 2005; Choi, Hecht, and Tayler 2013; Ferreira and Otley 2009; Rejc and Zaman 2012). In the academic context, one would thus expect a balance between research-related and teaching-related performance evaluation measures.

Bogt and Scapens (2012) report a lack of literature concerning performance measurement systems in universities. However, there is ample evidence that research is valued more highly than teaching (Gomez-Mejia and Balkin 1992; Katz 1973; Konrad and Pfeffer 1990; Newman 2008) and that many universities are placing increasing weight on research performance indicators for promotion, tenure, compensation and performance evaluation purposes (Arnold 2008; Bogt and Scapens 2012; Douglas 2013; Gendron 2008; Hopwood 2008; Long et al. 2009; Parker 2012; Sangster 2011).

In the USA, tangible evidence that research is valued more highly than teaching is provided by university compensation systems. Gomez-Mejia and Balkin (1992) found that in the USA the number of publications in top-tier journals was the main determinant of faculty pay. Notably they also investigated teaching effectiveness as a determinant of faculty pay and provide empirical evidence of a positive relationship between teaching effectiveness and pay, although this effect holds only for the segment of faculty with outstanding research records. A superior teaching performance is thus only reflected in a higher salary when a certain research quality threshold is reached. In contrast, Konrad and Pfeffer (1990) found no relationship between teaching performance and faculty salaries.

Tangible indicators reflecting the value of research in the European context are more diverse. Bogt and Scapens (2012) report that while quality research in the UK and the Netherlands may not elicit direct financial benefits, it impacts indirectly via faster promotion and job offers from other universities. In the UK, most universities require research excellence for the higher and more prestigious grades (such as professor) and do not consider applications for such posts based on teaching activities (Parker 2008). Sangster's (2011, 575) personal experience is consistent with Parker's finding: 'At all stages of my career since my first academic appointment, my promotion, salary rises and appointments have all been obtained largely on the basis of the quality of my research'. Further, many universities reward quality research with financial bonuses. As Olsson (2013) reports, at the BI Business School in Norway a top publication is rewarded with a 20,000 euro bonus; the school makes no mention of similar bonuses for high-quality teaching. Increasingly, many universities reward quality research with a smaller teaching load. Newman (2008) reports that academics at the University of Leeds publishing in top-tier journals receive 40 additional research days per year, while publishing in standard journals is rewarded with 5 additional

research days. If their research performance deteriorates, they are again penalised with higher teaching loads.

There is also less tangible evidence for the lower status of teaching in higher education institutions. Young (2006), for example, reports that academics perceived teaching as associated with low status and rewards of tenure and promotion as predominantly accruing in relation to research.

In any context, rational actors quickly adapt to performance measurement systems (Gendron 2008; Moya, Prior, and Rodriguez-Perez 2015). In the context of the increasing fixation on research performance indicators (Parker 2012), individual academics are increasingly content to provide merely satisfactory teaching to protect their research time (Bogt and Scapens 2012).

Research and teaching quality relationship hypotheses

The relationship between research and teaching performance is subjected to ongoing theoretical and empirical enquiry and controversy (Elton 2001; Grant and Wakelin 2009; Malcolm 2014). The polemic revolves around three main topics: the operationalisation and measurement of respective concepts, the nature of the relationship and the causality of the relationship.

Research performance in academia typically refers to scientific advancements, predominantly published in academic journals. A widely accepted convention holds that the best research achievements are published in the most prestigious scientific journals and have a high impact, while lower quality achievements are published in less prestigious journals and have a low impact (Harvey et al. 2010). It is noteworthy that, with respect to research, the same performance measures (e.g. publications, citations and grants) can be meaningfully applied at the organisational and individual levels.

Teaching performance relates to the number of students, degrees awarded and the quality of the education provided (Bogt and Scapens 2012). Yet these performance measures can only be applied at the organisational level, whereas individual instructors/teachers are responsible for the quality of education provided in the delivery of the courses they teach. In assessing their teaching quality, student experiences and perceptions also traditionally play an important role (Arnold 2008; Bedggood and Donovan 2012; Lindsay, Breen, and Jenkins 2002).

Both activities involve a quantity (productivity) and quality component. While quantity indicators for research (e.g. number of publications, citations and grants) and teaching (e.g. number of courses taught, hours in class and theses supervised) are directly observable, no directly observable indicators exist for the quality of research and teaching. In such instances, performance measurement systems often resort to surrogation, whereby a set of measures is derived to represent the unobserved construct of interest (Choi, Hecht, and Tayler 2013). Despite some discord in the academic community (Sangster 2011), the quality of research is usually assessed by its impact and measured by the number of papers published in an exclusive set of high-quality (top) journals or the number of citations (Harvey et al. 2010; Long et al. 2009; Stack 2003). Teaching quality/effectiveness is usually gauged via student assessments (Hattie and Marsh 1996). Although the use of student assessments is controversial (Balam and Shannon 2010; Bedggood and Donovan 2012), much of the literature in the area concludes that student assessments are generally consistent and valid (Arnold 2008; Liu 2012).

Concerning the nature of the relationship, Hattie and Marsh (1996) propose eight theories that posit three alternative relationships. The first group of theories, comprising the scarcity model, the differential personality model and the divergent reward system model, posits a negative relationship. The second group of theories, including the conventional wisdom model and the 'g' model, posits a positive relationship. Finally, the third group of theories, making up the different enterprises model, the unrelated personality model, and the bureaucratic funding model, hypothesises no relationship.

Empirical evidence supports the third. In their comprehensive meta-analysis Hattie and Marsh (1996) conclude that there is no relationship. It is worth noting that 86% of studies examined in their study used a productivity indicator for research performance (number of publications), whereas quality measures of research were used in fewer than 10% of the studies. For teaching performance, 90% of studies used student evaluations with the remainder being self and peer evaluations. More recent studies are also consistent with prior findings (e.g. Arnold 2008; Galbraith and Merrill 2012; Malcolm 2014; Marsh and Hattie 2002).

The third point of debate is the causality of the relationship. One line of argument in the literature is that research activity enhances teaching. As Lindsay, Breen, and Jenkins (2002) note, students are able to associate a lecturer's research activity with benefits such as knowledge currency, credibility, competence in supervision and enthusiasm (in the same way as they are able to identify drawbacks of a lecturer's research activity, such as reduced availability competition with teaching, and curriculum distortion). Research-led teaching is considered the preferred approach to teaching at many universities (Schapper and Mayson 2010). On the other hand, teaching can also enhance research. This comes through a variety of channels, such as honing understanding, learning through teaching, student questions and pedagogical experiments (Becker and Kennedy 2005). Despite long-standing research into the relationship most studies do not delve into the issue of causality. Rather, the most recent research phase focuses on how the research-teaching axis can enhance the quality and outcomes of the learning experience for both students and academics (Malcolm 2014).

While acknowledging alternative views regarding causality, our study models teaching quality as a dependent variable of research performance, on the basis of three arguments. First, many universities strive to facilitate research-led teaching due to an assumption of the positive effect of research on teaching (Schapper and Mayson 2010). Second, this assumption is implicitly embedded in contemporary performance measurement systems, which are prevalently research based (Parker 2012). Third, similar model specifications have recently been used by Arnold (2008) and Galbraith and Merrill (2012), where research activity variables were specified as predictors of teaching effectiveness.

An important weakness of prior research is the failure to adjust research performance for quality (Stack 2003). To make a more informed judgement of the effect of research activity on teaching quality, we divide research performance into productivity and quality components.

Research productivity refers to the amount of research output, regardless of its quality. Prior literature suggests that research activity bears both positive (e.g. knowledge currency) and negative (e.g. reduced time) implications for teaching (Lindsay, Breen, and Jenkins 2002). It is thus not surprising that theoretical support can be found for the differing interpretations of the relationship (Hattie and Marsh 1996). Performance evaluation systems providing incentives for research productivity can potentially moderate this relationship. However, trade-offs encountered by academics at the

input level (e.g. time allocated between research/teaching) are likely to be mediated by inverse trade-offs at the outcome level (e.g. knowledge currency/instructor's availability) (Arnold 2008).

The expected relationship hence follows prior empirical evidence (Galbraith and Merrill 2012; Marsh and Hattie 2002; Ramsden and Moses 1992), summarised in a comprehensive meta-analysis by Hattie and Marsh (1996) that these are unrelated enterprises. These prospects remain in the context of a research-biased performance evaluation system. The first argument for such projections is that bias towards research indicators is an enduring global phenomenon (Bogt and Scapens 2012; Parker 2012) hence implicitly embedded also in prior evidence. Second, as noted in some recent studies, the behaviour of academics is guided more by organisational culture (e.g. collegiality and autonomy) (Edgar and Geare 2013) and their intrinsic motivation (Kallio and Kallio 2014) than by managerial practices (e.g. performance appraisals). Thus, we posit the following hypothesis:

Hypothesis 1: Research productivity is not related to teaching quality.

Research quality refers to the impact rather than amount of research output. Prominent researchers who produce breakthrough knowledge often have strong personal traits, such as competence, diligence, motivation and orientation towards achievement (Allison and Stewart 1974; Edgar and Geare 2013; Kremer 1991; Woodburne 1952). A common characteristic of superior scholars is that they excel in all their activities (Kremer 1991; Woodburne 1952). Kallio and Kallio (2014) and Westergaard (1991) hence suggest that quality scholarship performance is a function of personal characteristics, irrespective of the activity undertaken. Therefore, performance evaluation systems providing incentives for quality research are congruent with quality teaching.

Despite relatively scarce empirical research, Hattie and Marsh (1996) noted that the few studies in their meta-analysis which used quality indicators for research performance uncovered a small positive relationship with teaching effectiveness. Further, Stack (2003) found a significant positive relationship between the quality of research, measured by citations, and student evaluations of teaching. In a more recent study, Stanton, Taylor, and Stanaland (2009) found that publishing in prestigious academic journals contributes significantly to positive student perceptions of teaching experience. Also of note is that Gomez-Mejia and Balkin (1992) find a positive relationship between teaching effectiveness and salary that is contingent upon outstanding research. Collectively, this suggests that excellent research is consistent with teaching quality and hence we posit the following hypothesis.

Hypothesis 2: Research quality is positively related to teaching quality.

Research design

The hypotheses are tested using archival data on research and teaching performance for a sample of academics across a wide range of faculties and scientific disciplines. A single university setting was adopted because we rely on archival collected data for teaching quality. A valid comparison in a multi-university setting would require the deployment of an identical measurement instrument for teaching quality at different universities, which is usually not the case.

Model specification

To examine the effect of research productivity and research quality on an academic's teaching quality, we devised the following regression model, including four control variables:

$$\text{Teaching quality} = a + b_1 * \text{Research productivity} + b_2 * \text{Research quality} + b_3 * \text{School research culture} + b_4 * \text{Number of courses taught} + b_5 * \text{Number of students taught} + b_6 * \text{Assistant professor} + b_7 * \text{Associate professor} + b_8 * \text{Full professor} + e.$$

School research culture was included due to prior evidence that organisational culture is an important facilitator of academic performance (Edgar and Geare 2013). It is common within universities that some schools/departments put more emphasis on research and others put more emphasis on teaching (Allison and Stewart 1974; Arnold 2008) which in effect determines how the academics within these organisations value and prioritise their work. We posit that school culture oriented towards research outcomes is inversely related to teaching quality.

Number of courses and students taught variables were included as indicators of teaching quantity. A positive relationship between the extent of teaching experience and teaching quality can be expected; however the causality of this relationship is less obvious. On the one hand, greater teaching experience may improve teaching quality. On the other hand, higher quality teachers may be assigned more courses than lower quality teachers (Ragan and Walia 2010). Number of students taught was included due to enduring evidence that the relationship between teaching quality and the number of students taught is negative (Arnold 2008; Ragan and Walia 2010; Stack 2003), revealing that student satisfaction is negatively affected by mass education.

The fourth control variable is academic title (i.e. teaching assistant, assistant professor, associate professor and full professor), modelled as a set of three dummy variables. This variable also reflects the length of an academic's teaching (and research) experience. The general learning curve theory suggests that senior academics with longer teaching experience are more effective teachers. On the other hand, senior academics may be more involved in administrative services (Bentley and Kyvik 2012) and/or less familiar with contemporary teaching methods (Liu 2012), suggesting a negative effect.

The model was tested using two sets of data. The first set contained original archival data for teaching and research performance administered by the case university. However, the use of original data across a heterogeneous sample of academics poses comparability concerns. This is especially pertinent to research indicators because researchers from different scientific disciplines exhibit different research orientations, research products and co-authorship patterns (Allison and Stewart 1974; Sabharwal 2013). As Hattie and Marsh (1996) note, it makes more sense to benchmark academic performance against peers within the same discipline rather than against peers from completely different disciplines. For this purpose, the research productivity and teaching quality of individual researchers was benchmarked relative to the peers affiliated to the same faculty.

Case university performance evaluation context

The case university is the University of Maribor, Slovenia (EU). This university is an ideal setting for our investigation for two reasons. First, unlike most other universities, it publicly reveals the student evaluation scores of its academic staff. Second, in

Slovenia academic staff turnover is very low (Bartolj and Polanec 2012; Obadic and Aristovnik 2011). Most academics start their careers as teaching assistants and continue their careers at the same faculty until retirement (Cadez 2013). In effect, the data reflect research and teaching performance objectively across a longer time span.

The University of Maribor is the second largest, oldest and most prestigious of the three (public) universities in Slovenia, after the University of Ljubljana. There are around 1800 total staff (of whom about two-thirds are academics) who cater for some 21,000 students. The vast majority of staff and students are Slovenian nationals, although the university is increasingly internationalising its activities (Rebolj 2011). Like all public universities in the country, it receives most of its funding from the Ministry of Education, based primarily on student programme offerings and student numbers although a small proportion of funding is also based also on faculty research performance. These funds, provided by the Slovenian Research Agency, are competitive and merit based. As noted by Auranen and Nieminen (2010), similar input-oriented funding systems are prevalent in continental Europe.

The university's mission is to expand and transfer knowledge in the field of science, art and education, while its vision is to become a globally recognised innovation ecosystem, inspiring the creativity of both employees and students. The university currently comprises 17 faculties (schools) covering all scientific disciplines. Table 1 provides data on the constituent faculties and their academic staff. The academic career track at the University of Maribor (and also other Slovenian universities) involves four grades, that is, teaching assistant, assistant professor, associate professor and full professor (the position of lecturer is also common but can be seen as a deviation from an academic career track for staff focusing exclusively on teaching).

Uniform across the university, the performance evaluation system is highly quantitative and chiefly used for promotion purposes. Each academic is eligible to apply for a promotion in academic grade every five years. Performance evaluations for promotion purposes are therefore periodic (every five years), except for full professors who are no longer subject to performance evaluation for promotion purposes.

Despite the fact that the university's mission statement embraces knowledge transfer (i.e. teaching) and knowledge expansion (i.e. research) equally, evaluation criteria are predominantly research based. One's entire research and teaching portfolio is evaluated (e.g. number of monographs, journal articles, conference presentations, citations, research grants, professional articles, doctoral students supervisions, masters students supervisions, student textbooks written, student assessments, etc.) by a sophisticated measurement system, however the most important selection criterion for promotion is internationally relevant publications (i.e. a minimum of 3 for an assistant professor, a minimum of 6 for an associate professor and a minimum of 12 for a full professor). International relevance is interpreted in the university promotion criteria as a publication in a journal included in the Web of Science Core Collection, irrespective of the journal impact factor (Cadez 2013). Although teaching effectiveness is one of the criteria, very few staff fail to meet this criterion for promotion purposes as only a minimum satisfactory standard is required (Obadic and Aristovnik 2011).

In short, the performance evaluation system at the University of Maribor is predominantly research based. More precisely, the system provides strong incentives for research productivity. Research quality is a less selective norm, as only international relevance, not excellence, is required. The relevance of teaching quality in performance evaluation is marginal.

Table 1. Constituent faculties of the University of Maribor.

Faculty	Acronym	Scientific discipline	Number of academic staff ^a
1. Natural Sciences and Mathematics	FNSM	Natural	75
2. Chemistry and Chemical Engineering	FCCE	Natural	42
3. Civil Engineering (Construction)	FCE	Technology	118
4. Energy Technology	FET	Technology	23
5. Electrical Engineering and Computer Science	FEECS	Technology	167
6. Mechanical Engineering	FME	Technology	83
7. Medicine	FM	Life	62
8. Health Sciences	FHS	Life	27
9. Agriculture and Life Sciences	FALS	Life	67
10. Education	FE	Social	123
11. Economics and Business	FEB	Social	72
12. Organisational Sciences	FOS	Social	55
13. Law	FLA	Social	26
14. Arts	FA	Arts/humanities	145
15. Logistics	FLO	Interdisciplinary	45
16. Tourism	FT	Social	Not available
17. Criminal, Justice and Security	FCJS	Social	Not available
Total			1130

^aStaff numbers were derived from the University of Maribor's teaching performance database publicly available at <http://www.uni-mb.si/podrocje.aspx?id=439> (downloaded in May 2012).

Source: Rebolj (2011).

Sample selection

The distribution of staff in terms of academic titles is provided in Table 2. The table includes staff from 15 faculties only as data were not available for staff from the Tourism and the Criminal, Justice and Security Faculties. As evident from Table 2, 62.2% of staff hold one of the three professor titles and 71.9% of staff hold a PhD (a PhD is a prerequisite for any professor title). The total population with available teaching performance data at the University of Maribor is 1130 academic staff. Since our aim is to test the relationship between teaching quality and research performance, the overall population was filtered to screen out academic staff not actively engaged in research and staff with potentially invalid teaching performance data.

The first filter deployed to screen out non-active research staff is the holding of a PhD title. We presume that holding a PhD is a proxy indicator that the title holder has been engaged in research activities in the past and is likely to exhibit some tangible research output (i.e. publications). With this filter, the sample was reduced to 812 academic staff.

In the interest of the results' validity, a second filter excluded all research staff with potentially invalid and unreliable teaching performance data. Since teaching quality was appraised using student assessments, we excluded all staff where the number of completed surveys did not meet an acceptable threshold, set at 30 completed student surveys per researcher. This filter further reduced the sample to 620 academic staff.

Table 2. Academic staff distribution in terms of academic title.

Academic title	Number	in %
Full professor	208	18.4
Associate professor	183	16.2
Assistant professor	312	27.6
Teaching assistant with doctorate/doctor	109	9.6
Teaching assistant without doctorate	148	13.1
Lecturer (without doctorate)	89	7.9
Other academic staff*	81	7.2
Total	1130	100

*Manly includes foreign language teachers and staff without an academic title.

Source: University of Maribor teaching performance database.

Variable measurement

Teaching quality was gauged using student assessments administered by the University of Maribor for the 2010–2011 academic year. The questionnaire is uniform across all faculties. Measured on a five-point Likert scale ranging from –2 to +2 it comprises eight dimensions of teaching effectiveness: (1) interest and comprehensibility; (2) cooperation with students; (3) punctuality and availability; (4) selection of literature and materials for the course; (5) overall teaching quality; (6) aptness of student evaluation methods; (7) objectivity in student marking; and (8) feedback to students after marking. Similar to Marsh and Hattie (2002), we find that the indicators are highly correlated. All 28 partial bivariate correlations exceed the value of 0.8. It should be noted that, although course instructors are evaluated at the course level, the university publicly discloses teaching performance data at the instructor level only, not the course level. The mean score from all eight items is also provided as a measure of an instructor's overall teaching effectiveness. We used this overall mean score as a proxy for teaching quality. Benchmarked teaching quality (for model 2) was calculated by dividing each academic's overall score by the mean overall score for all academics in the faculty.

Research productivity was assessed using the method of publication count. The measure deployed is the number of publications in journals included in the Web of Science Core Collection database in the period 2006–2011. While acknowledging that research output can also include other outlets, such as monographs, book chapters and publications in journals not included in the Web of Science, our selection rests on the following arguments. First, the Web of Science database comprises peer-reviewed journals implying that the published research meets acceptable international quality standards (Sabharwal 2013). Second, the University of Maribor's promotion criteria explicitly require publications in these journals to secure promotions; hence we presume that researchers mainly target these journals as their research outlets. Third, the use of a single database reduced the subjectivity concerned with the assessment of an individual's research output (Auranen and Nieminen 2010). Benchmarked research productivity (for model 2) was calculated by dividing each academic's publication count by the average number of publications per academic within the same faculty.

Research quality was measured as the proportion of papers published in high-quality journals relative to all papers published by a particular researcher. As a proxy for journal quality, we use the impact factors of journals in the Web of Science database. Since impact factors in the Web of Science database vary widely across subject fields, we consider the relative rankings of journals within a particular

subject field instead. Following Cadez (2013) and Harvey et al. (2010), we label journals in the top quartile of their respective subject fields as high-quality journals. This measure may not necessarily be the most objective measure of research quality but we base our selection on the fact that, first, a relative measure excludes a productivity component and, second, the same measure is also used by the Slovenian Research Agency for evaluating research quality of Slovenian researchers (Cadez 2013).

School research culture was gauged by the amount of external research funding received from the Slovenian Research Agency. This funding is competitive and merit based and hence we take the success in securing these funds as a proxy for the research culture of a school. To eliminate annual fluctuations in research funding we have taken a three-year average. To eliminate the effect of school size, we calculated the amount of research funds per researcher.

Number of courses taught is the number of different courses an academic taught in the 2010–2011 academic year.

Number of students is the total number of students who attended courses taught by a particular academic in the 2010–2011 academic year.

Academic title is the title an academic held in the 2010–2011 academic year. For the purpose of quantitative analysis, this variable is represented in the model by three dummy variables (assistant professor, associate professor and full professor).

Data collection and analysis

The teaching performance data were available in a Microsoft Excel Worksheet administered by the University of Maribor. The worksheet contained data for 15 variables for 1130 academics for the 2010–2011 academic year. The variables included were faculty affiliation, name and surname, academic title, number of courses taught, number of students attending the courses taught by the academic, number of students completing the survey, scores for each of the eight dimensions of teaching quality and the mean teaching quality score.

Research performance data were collected manually for each academic from the SICRIS database (a Slovenian research information system), which summarises integrated data directly from the Web of Science database. The period appraised was from 2006 to 2011.

The model was tested with multiple linear regression analysis. An important consideration in the regression analysis is the handling of outliers (Freund, Wilson, and Sa 2006; Hair et al. 2006). Teaching quality, the dependent variable in the model, measured using a five-point Likert scale, is almost normally distributed with no potential outliers. The distribution of research productivity variable is substantially skewed, consistent with the long-established Lotka's law (Seglen 1992) that in a typical case 6% of publishing academics produce 50% of all papers (Allison and Stewart 1974). The distribution of research quality variable is fairly uniform. Since outliers cannot be categorically characterised as either beneficial or problematic, Hair et al. (2006) suggest a substantive analysis of respective observations. Both extremely research-productive (Allison and Stewart 1974; Seglen 1992) as well as high- and low-quality-oriented researchers (Edgar and Geare 2013; Harvey et al. 2010) have long been represented in the academic community. Therefore, the decision has been taken to keep all values in the sample.

Other assumptions of regression analysis were also considered. Non-normal distribution of independent variables is not seen as problematic since the large sample size

effectively diminishes its impact (Hair et al. 2006). Correlation between independent variables is not high, suggesting little concern for multicollinearity (Freund, Wilson, and Sa 2006). Visual inspection of partial bivariate scatterplots suggests no potential heteroscedasticity problems (Hair et al. 2006).

The two models (one with original data, one with data benchmarked against peers) yielded fairly similar regression parameters. Significantly, visual inspection of the residuals plots, with residuals scattered symmetrically around the theoretical value, suggests little concern for non-normality, nonlinearity and heteroscedasticity (Tabachnick and Fidell 2007).

Findings

Table 3 provides descriptive statistics for the whole sample (see the last row) and each faculty separately. On average, each academic taught 4.87 courses, which were attended by 178 students (i.e. 36.5 students per course). The mean teaching quality score across all faculties is 1.28, the mean number of papers published in the period examined is 7.34 and the mean number of papers published in top-quartile journals is 2.02 papers or 27.5%.

The teaching-related data exhibit a degree of variability across faculties. The mean number of courses taught ranges from 1.71 (law) to 7.77 (mechanical engineering). The mean number of students taught ranges from 94 to 350. Mean teaching quality scores range from 1.06 (agriculture and life sciences) to 1.46 (natural sciences and mathematics). It is noteworthy that the vast majority of academics scored above the midpoint of the scale.

Consistent with expectations, the variation in research productivity is considerably greater than the variation in teaching quality scores. Mean research productivity ranges from 0.86 (law) to 26.06 chemistry and chemical engineering (papers). The number and proportion of papers published in top-quartile journals also varies substantially across faculties from 0% (law) to 62.9% (chemistry and chemical engineering).

Results of the two regression models are presented in Table 4. The first model presents original data and the second data benchmarked against peers affiliated with the same faculty. As is evident, the estimated parameters in the two models are fairly similar. Hypothesis 1 is supported in both models – research productivity is not related to teaching quality. Hypothesis 2 is also supported in both models – research quality is positively related to teaching quality. With respect to control variables, school research culture is negatively related, while number of courses taught by an academic is positively related to teaching quality. The number of students taught does not influence their perceptions of the instructor's quality of teaching. Finally, teaching quality of academics holding a full professor title is significantly lower compared to their colleagues in a teaching assistant role with a PhD title. The variance explained by the models is small, consistent with other recent studies in the area (Arnold 2008; Galbraith and Merrill 2012).

Discussion and conclusion

This study contributes to the ongoing debate in higher education as to whether active involvement in research is essential to be a good university teacher. Unlike most prior studies in the area, it explicitly considers the moderating effect of a performance evaluation system. The hypotheses are tested in the context of a university where the

Table 3. Research and teaching descriptive statistics.

Faculty ^a	Number of academic staff holding a PhD	Teaching performance score ^b			Research productivity score			RQ ^c Mean	RQ% ^d Mean	Research funds per capita in € Mean	Number of courses Mean	Number of students Mean
		Mean	Min	Max	Mean	Min	Max					
FNSM	36	1.46	0.75	1.95	12.86	0	88	2.56	19.9%	6845	4.97	125
FCCE	32	1.26	0.17	1.88	26.06	0	246	16.38	62.9%	29901	4.94	112
FCE	49	1.06	-0.12	1.73	3.73	0	13	1.22	32.7%	2724	4.00	139
FET	12	1.21	0.60	1.50	17.17	0	62	2.75	16.0%	136	5.08	170
FEECS	103	1.21	-0.37	1.99	8.09	0	69	1.18	14.6%	15583	6.13	156
FME	65	1.28	0.34	1.97	11.20	0	46	2.98	26.6%	33021	7.77	198
FM	37	1.28	0.21	1.88	10.30	0	51	2.03	19.7%	7479	2.19	119
FHS	14	1.37	0.41	1.80	9.50	1	28	2.29	24.1%	2687	3.14	204
FALS	27	1.41	0.10	1.88	7.89	0	31	1.33	16.9%	5708	6.07	94
FE	32	1.34	-0.06	1.80	4.34	0	33	1.00	23.0%	500	3.34	222
FEB	49	1.33	0.71	1.85	4.02	0	21	0.57	14.2%	3690	3.43	231
FOS	36	1.32	0.72	1.69	1.78	0	7	0.14	7.9%	2944	3.81	170
FLA	14	1.24	0.66	1.72	0.86	0	3	0.00	0.0%	7194	1.71	350
FA	97	1.30	0.12	1.92	1.15	0	15	0.15	13.0%	4975	5.20	224
FLO	17	1.24	0.48	1.75	3.12	0	15	0.24	7.7%	30	3.71	198
Total	620	1.28	-0.37	1.99	7.34	0	246	2.02	27.5%	10395	4.87	178

^aSee Table 1 for an explanation of the faculty acronyms.^bScale range from '-2' to '2'.^cRQ – mean number of papers published in journals ranked in the top quartile of a respective subject field.^dRQ% – proportion of papers published in journals ranked in the top quartile of a respective subject field.

Table 4. Regression analysis parameters.

Parameter	Model 1 – original data	Model 2 – benchmarked data
Observations	620	620
Intercept	1.277**	0.995**
Research productivity	0.000	0.001
Research quality	0.002*	0.003*
School research culture	−0.00004*	−0.00003*
Number of courses taught	0.018**	0.016**
Number of students taught	0.000	0.000
Assistant professor	−0.032	−0.031
Associate professor	−0.024	−0.023
Full professor	−0.115*	−0.109*
F value	3.32**	4.31**
R ²	0.04	0.06

*The coefficient is significant at the $p < .05$ level (two tail).

**The coefficient is significant at the $p < .01$ level (two tail).

performance evaluation system provides strong incentives for research productivity, an increasingly common occurrence in contemporary academia (Bogt and Scapens 2012; Gendron 2008; Moya, Prior, and Rodriguez-Perez 2015; Parker 2008). In addition, the study tackles an important drawback of prior studies, that is, the failure to adjust research output for its quality (Stack 2003). By breaking down research performance into productivity and quality components, this study contributes to our understanding of the research–teaching relationship in higher education.

Consistent with hypothesis 1 and ample prior evidence we find that research productivity (the amount of research output) is not related to teaching quality. Alternative explanations of this phenomenon are discussed in a number of prior studies (e.g. Arnold 2008; Galbraith and Merrill 2012; Hattie and Marsh 1996; Kremer 1990; Marsh and Hattie 2002; Westergaard 1991) and will not be repeated here. The heart of the problem is that two separate skill sets are integrated into a single job. Researchers are asked to teach and teachers are asked to do research, even though these two capabilities appear to have little to do with each other.

We find, however, consistent with hypothesis 2, that research quality (the impact of research output) is positively related to teaching quality. This finding is consistent with some recent studies in the area, indicating that publishing in prestigious journals (Stanton, Taylor, and Stanaland 2009) and the number of citations (Stack 2003) affect teaching quality positively. As Lindsay, Breen, and Jenkins (2002) note, students are able to associate benefits (knowledge currency, credibility, etc.) and drawbacks (reduced availability, competition with teaching, etc.) with a lecturer's research activity. It appears though that the benefits outweigh the drawbacks only when academics conduct high-quality research. In other words, it is not the publication count but the quality of what is being offered that in effect enhances teaching quality (Hopwood 2008; Lindsay, Breen, and Jenkins 2002).

It is meaningful that a positive relationship between research quality and teaching quality was detected in a performance evaluation context that provides strong incentives for research productivity but few incentives for research quality and teaching quality. This finding is consistent with long-standing claims that superior scholars excel in any activity they perform (Kremer 1991; Woodburne 1952). The behaviour of these academics is guided more by their intrinsic motivation to produce and

disseminate knowledge at high standards of excellence (Deci 1971; Godin and Gingras 2000; Westergaard 1991) than by managerial practices that evaluate their work (Kallio and Kallio 2014). This is an encouraging finding in the contemporary academic setting that often focuses on maximising production regardless of the quality of the product (Kallio and Kallio 2014). Motivational theories from psychology, such as Ajzen and Fishbein's (1977) attitude-behaviour theory or Ryan and Deci's (2000) self-determination theory, seem useful to explain the phenomenon; however we leave this interesting question for further enquiry.

Results related to control variables in the regression model also merit interpretation. First, as expected, the research orientation of a particular school (measured by their success in attracting external merit-based research funding) is negatively related with teaching quality. This is consistent with prior findings that the behaviour of academics is guided also by organisational culture (Edgar and Geare 2013), which affects how individuals value and prioritise their work. It appears that in the case university securing external research funding is a higher priority than quality teaching.

Second, as predicted, the number of courses taught is positively related to teaching quality. While this suggests that diversity in teaching experience enhances teaching quality, it is also possible that more effective teachers are assigned a higher teaching load. The causality of this relationship is thus unclear. Contrary to predictions, the number of students taught is not related to teaching quality. However, this study investigated the relationship at the instructor level, not course level, which renders our analysis incomparable with prior studies in the area (e.g. Arnold 2008; Stack 2003). It is, for instance, possible that the same instructor was evaluated highly for the delivery of a small (in terms of students) course and lowly for the delivery of a large course.

Finally, holding a full professor title impairs the quality of students' learning experience. A number of alternative explanations seem plausible for this finding. First, the contemporary shift towards more interactive teaching methods supported by new technology options may be better mastered by junior staff (Balam and Shannon 2010); in the recent investigation of a distance learning context, full professors ranked the lowest in terms of teaching effectiveness (Liu 2012). Second, senior academics are generally more involved in administrative services, thus leaving them less time for quality teaching (and research) (Bentley and Kyvik 2012). Third, having obtained the highest academic rank, their external motivation for high performance may deteriorate (Kallio and Kallio 2014; Ryan and Deci 2000) or they may have focused on one particular area (Kremer 1991).

This study also contributes to the performance management literature. The case university is an example of a 'modern' university (Bogt and Scapens 2012), where strong incentives are provided for research productivity (though not quality). Somewhat paradoxically, research quality with an observed positive effect on teaching quality is not really stimulated by the current performance evaluation system. While these findings may imply that the current productivity-oriented performance evaluation system at the case university is somewhat dysfunctional, it is not inconsistent with the organisational strategy. As evident from the vision statement of the case university, it strives for global recognition, not global excellence. In this context, the performance management system that fosters internationally relevant publications (and in effect does not impair teaching) seems aligned with the organisational strategy to secure its effective implementation (Cadez and Guilding 2008; Chenhall 2005; Ferreira and Otley 2009).

From a holistic perspective, our findings corroborate prior propositions that quality scholarship performance is more a function of individual motivations than the work environment (Edgar and Geare 2013; Kallio and Kallio 2014; Westergaard 1991). The findings also discount fears that research-based performance evaluation in academia may be detrimental to quality teaching (Arnold 2008; Bogt and Scapens 2012; Gendron 2008). In fact, performance evaluation systems providing incentives for high research quality appear congruent with quality teaching.

In interpreting the study's findings, some limitations should be borne in mind. As noted before, opinions about using publications and student assessments as proxies for research and teaching quality are controversial, although these measures remain widely used (Galbraith and Merrill 2012; Liu 2012). Article counts do not necessarily reflect the final aim of research activity, that is, knowledge production (Butler 2002); student evaluations do not necessarily reflect the final aim of teaching, that is, student learning (Bedggood and Donovan 2012). Alternative measures of teaching quality, proposed in the recent literature, include student learning outcomes (Bedggood and Donovan 2012; Galbraith and Merrill 2012; Malcolm 2014) or their career advancement after completing the programme (Bartolj and Polanec 2012; Gendron 2008). Further, although comparable with recent prior studies (Arnold 2008; Galbraith and Merrill 2012), the explained variance in our model is relatively small. Other factors, such as the academic's ability, motivation and time for teaching, appear to be more important predictors of teaching quality than research activity (Boex 2000). Nevertheless, the aim of this study is not to maximise the explanatory power of the model but to yield consistent parameter estimates. While a potential for omitted variable bias is present in any model, an ordinary least square estimation method yields consistent parameter estimates if the omitted variables are not correlated with the included regressors (Cameron and Trivedi 2005).

Some limitations are also study specific. First, a weighted number of publications (in terms of authors) may be a better indicator of an individual's productivity. Also, even though the same measure is used by the Slovenian Research Agency to evaluate research quality (Cadez 2013), some may justifiably argue that our measure of research quality is not the most objective. Despite these weaknesses, this study provides novel insights into the relationship between research, teaching and performance evaluation in the contemporary academic environment.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

- Ajzen, I., and M. Fishbein. 1977. "Attitude-behavior Relations: A Theoretical Analysis and Review of Empirical Literature." *Psychological Bulletin* 84: 888–918.
- Allison, P. D., and J. A. Stewart. 1974. "Productivity Differences Among Scientists: Evidence for Accumulative Advantage." *American Sociological Review* 39 (4): 596–606.
- Arnold, I. J. M. 2008. "Course Level and the Relationship Between Research Productivity and Teaching Effectiveness." *Journal of Economic Education* 39 (4): 307–21.
- Auranen, O., and M. Nieminen. 2010. "University Research Funding and Publication Performance – An International Comparison." *Research Policy* 39: 822–34.
- Balam, E., and D. Shannon. 2010. "Student Ratings of College Teaching: A Comparison of Faculty and Their Students." *Assessment & Evaluation in Higher Education* 35 (2): 209–21.

- Bartolj, T. and S. Polanec. 2012. "College Major Choice and Ability: Why is General Ability not Enough?" *Economics of Education Review* 31 (1): 996–1016.
- Becker, W., and P. Kennedy. 2005. "Does Teaching Enhance Research in Economics?" *American Economic Review* 95 (2): 172–76.
- Bedggood, R., and J. Donovan. 2012. "University Performance Evaluations: What are We Really Measuring?" *Studies in Higher Education* 37 (7): 825–42.
- Bentley, P. J., and S. Kyvik. 2012. "Academic Work from a Comparative Perspective: A Survey of Faculty Working Time Across 13 Countries." *Higher Education* 63 (4): 529–47.
- Besancenot, D., J. R. Faria, and R. Vranceanu. 2009. "Why Business Schools do so Much Research: A Signalling Explanation." *Research Policy* 38: 1093–1101.
- Boex, L. F. J. 2000. "Attributes of Effective Economics Instructors." *Journal of Economic Education* 31 (3): 211–27.
- Bogt, H. J., and R. W. Scapens. 2012. "Performance Management in Universities: Effects of the Transition to More Quantitative Measurement Systems." *European Accounting Review* 21 (3): 451–97.
- Butler, L. 2002. "A List of Published Papers is no Measure of Value." *Nature* 419: 877.
- Cadez, S. 2013. "Social Change, Institutional Pressures and Knowledge Creation: A Bibliometric Analysis." *Expert Systems with Applications* 40 (17): 6885–93.
- Cadez, S., and C. Guilding. 2008. "An Exploratory Investigation of an Integrated Contingency Model of Strategic Management Accounting." *Accounting, Organizations and Society* 33 (7/8): 836–63.
- Cameron, A. C., and P. K. Trivedi. 2005. *Microeconometrics: Methods and Applications*. Cambridge: Cambridge University Press.
- Chenhall, R. H. 2005. "Integrative Strategic Performance Measurement Systems, Strategic Alignment of Manufacturing, Learning and Strategic Outcomes: An Exploratory Study." *Accounting, Organizations and Society* 30 (5): 395–422.
- Choi, J., G. Hecht, and W. Tayler. 2013. "Strategy Selection, Surrogation, and Strategic Performance Measurement Systems." *Journal of Accounting Research* 51 (1): 105–33.
- Deci, E. L. 1971. "Effects of Externally Mediated Rewards on Intrinsic Motivation." *Journal of Personality and Social Psychology* 18 (1): 105–15.
- Douglas, A. S. 2013. "Advice from the Professors in a University Social Sciences Department on the Teaching-Research Nexus." *Teaching in Higher Education* 18 (4): 377–88.
- Edgar, F., and A. Geare. 2013. "Factors Influencing University Research Performance." *Studies in Higher Education* 38 (5): 774–92.
- Elton, L. 2001. "Research and Teaching: Conditions for a Positive Link." *Teaching in Higher Education* 6 (1): 43–56.
- Ferreira, A., and D. Otley. 2009. "The Design and Use of Performance Management Systems: An Extended Framework for Analysis." *Management Accounting Research* 20 (4): 263–82.
- Freund, R., W. Wilson, and P. Sa. 2006. *Regression Analysis*. 2nd ed. Amsterdam: Elsevier.
- Galbraith, C., and G. Merrill. 2012. "Faculty Research Productivity and Standardized Student Learning Outcomes in A University Teaching Environment: A Bayesian Analysis of Relationships." *Studies in Higher Education* 37 (4): 469–80.
- Gendron, Y. 2008. "Constituting the Academic Performer: The Spectre of Superficiality and Stagnation in Academia." *European Accounting Review* 17 (1): 97–127.
- Godin, B., and Y. Gingras. 2000. "The Place of Universities in the System of Knowledge Production." *Research Policy* 29 (2): 273–78.
- Gomez-Mejia, L. R., and D. B. Balkin. 1992. "Determinants of Faculty Pay: An Agency Theory Perspective." *Academy of Management Journal* 35 (5): 921–55.
- Grant, K., and S. J. Wakelin. 2009. "Re-conceptualisation the Concept of a Nexus? A Survey of 12 Scottish IS/IM Academics' Perceptions of A Nexus Between Teaching, Research, Scholarship and Consultancy." *Teaching in Higher Education* 14 (2): 133–46.
- Hair, J. F. et al. 2006. *Multivariate Data Analysis*. 6th ed. New Jersey: Pearson/Prentice Hall.
- Harvey, C., A. Kelly, H. Morris, and M. Rowlinson. 2010. *Academic Journal Quality Guide*. London: The Association of Business Schools.
- Hattie, J., and H. W. Marsh. 1996. "The Relationship Between Research and Teaching: A Meta-Analysis." *Review of Educational Research* 66 (4): 507–42.
- Hopwood, A. G. 2008. "Changing Pressures on the Research Process: On Trying to Research in an Age When Curiosity is Not Enough." *European Accounting Review* 17 (1): 87–96.

- Kallio, K., and T. Kallio. 2014. "Management by Results and Performance Measurement in Universities – Implications for Work Motivation." *Studies in Higher Education* 39 (4): 574–89.
- Katz, D. A. 1973. "Faculty Salaries, Promotions and Productivity at a Large University." *American Economic Review* 63 (3): 469–77.
- Konrad, A. M., and J. Pfeffer. 1990. "Do You Get What You Deserve? Factors Affecting the Relationship Between Productivity and Pay." *Administrative Science Quarterly* 35 (2): 258–85.
- Kremer, J. 1990. "Construct Validity of Multiple Measures in Teaching, Research and Services and Reliability of Peer Ratings." *Journal of Educational Psychology* 82 (2): 213–18.
- Kremer, J. 1991. "Identifying Faculty Types Using Peer Ratings of Teaching, Research and Service." *Research in Higher Education* 32 (4): 351–61.
- Lindsay, R., R. Breen, and A. Jenkins. 2002. "Academic Research and Teaching Quality: The Views of Undergraduate and Postgraduate Students." *Studies in Higher Education* 27 (3): 309–27.
- Liu, O. 2012. "Student Evaluation of Instruction: In the New Paradigm of Distance Education." *Research in Higher Education* 53 (4): 471–86.
- Long, R., A. Crawford, M. White, and K. Davis. 2009. "Determinants of Faculty Research Productivity in Information Systems: An Empirical Analysis of the Impact of Academic Origin and Academic Affiliation." *Scientometrics* 78 (2): 231–60.
- Malcolm, M. 2014. "A Critical Evaluation of Recent Progress in Understanding the Role of the Research-Teaching Link in Higher Education." *Higher Education* 67 (3): 289–301.
- Marsh, H. W., and J. Hattie. 2002. "The Relation Between Research Productivity and Teaching Effectiveness – Complementary, Antagonistic, or Independent Constructs?" *The Journal of Higher Education* 73 (5): 603–41.
- Modell, S. 2003. "Goals Versus Institutions: The Development of Performance Measurement in the Swedish University Sector." *Management Accounting Research* 14 (4): 333–59.
- Moed, H. 2008. "UK Research Assessment Exercises: Informed Judgments on Research Quality or Quantity." *Scientometrics* 74 (1): 153–61.
- Moll, J., and Z. Hoque. 2011. "Budgeting for Legitimacy: the Case of an Australian University." *Accounting, Organizations and Society* 36 (2): 86–101.
- Moya, S., D. Prior, and G. Rodriguez-Perez. 2015. "Performance-based Incentives and the Behaviour of Accounting Academics: Responding to Changes." *Accounting Education: An International Journal* 24 (3): 208–32.
- Newman, J. H. 1907. *The Idea of a University*. London: Green & Co.
- Newman, M. 2008. "Leeds Rewards Publication in Top Journals with Bonus Research Time." *Times Higher Education*, May 1, 2008.
- Obadic, A., and A. Aristovnik. 2011. "Relative Efficiency of Higher Education in Croatia and Slovenia: an International Comparison." *Amfiteatru Economic* 13 (30): 362–76.
- Olsson, U. 2013. "Assessing Research Quality at Individual Level: The Case of BI Norwegian Business School." Paper presented at the EIASM conference "Quality in (Management) Research", Copenhagen.
- Parker, J. 2008. "Comparing Research and Teaching in University Promotion Criteria." *Higher Education Quarterly* 62 (3): 237–51.
- Parker, L. D. 2012. "Beyond the Ticket and the Brand: Imaging an Accounting Research Future." *Accounting and Finance* 52 (4): 1153–82.
- Ragan, J. F., and B. Walia. 2010. "Differences in Student Evaluations of Principles and Other Economics Courses and the Allocation of Faculty Across Courses." *The Journal of Economic Education* 41 (4): 335–52.
- Ramsden, P., and I. Moses. 1992. "Associations Between Research and Teaching in Australian Higher Education." *Higher Education* 23 (3): 273–95.
- Rebolj, D. 2011. *University of Maribor*. Maribor: University of Maribor.
- Reidpath, D., and P. Allotey. 2010. "Can National Research Assessment Exercises be Used Locally to Inform Research Strategy Development? The Description of a Methodological Approach to the UK RAE 2008 Results with a Focus on one Institution." *Higher Education* 59 (6): 785–97.
- Rejc, B. A., and Zaman, G. M. 2012. "Contemporary Performance Measurement Systems in Central and Eastern Europe." *Journal for East European Management Studies* 17 (1): 69–103.

- Robbins, L. 1963. *Report of the Committee on Higher Education*. London: HMSO.
- Rodman, K., R. Biloslavo, and S. Bratoz. 2013. "Institutional Quality of a Higher Education Institution from the Perspective of Employers." *Minerva* 51 (1): 71–92.
- Ryan, R. M., and E. L. Deci. 2000. "Self-determination Theory and the Facilitation of Intrinsic Motivation, Social Development, and Well-Being." *American Psychologist* 55 (1): 68–78.
- Sabharwal, M. 2013. "Comparing Research Productivity Across Disciplines and Career Stages." *Journal of Comparative Policy Analysis* 15 (2): 141–63.
- Sangster, A. 2011. "The ABS Journal Quality Guide: A Personal View." *Accounting Education: An International Journal* 20 (6): 575–80.
- Schapper, J., and S. Mayson. 2010. "Research-led Teaching. Moving From a Fractured Engagement to a Marriage of Convenience." *Higher Education Research and Development* 29 (6): 641–51.
- Seglen, P. 1992. "The Skewness of Science." *Journal of American Society for Information Science* 43 (9): 628–38.
- Stack, S. 2003. "Research Productivity and Student Evaluation of Teaching in Social Science Classes: A Research Note." *Research in Higher Education* 44 (5): 539–56.
- Stanton, A. A., R. L. Taylor, and A. J. S. Stanaland. 2009. "An Examination of the Relationship Between Research Attitudes and Behaviors of Business School Faculty." *Academy of Educational Leadership Journal* 13 (3): 37–49.
- Tabachnick, B., and L. Fidell. 2007. *Using Multivariate Statistics*. 5th ed. Boston: Pearson.
- Taylor, J. 2007. "The Teaching: Research Nexus: A Model for Institutional Management." *Higher Education* 54 (6): 867–84.
- Westergaard, J. 1991. "Scholarship, Research and Teaching: A View from the Social Sciences." *Studies in Higher Education* 16 (1): 23–28.
- Woodburne, L. S. 1952. "The Qualifications of Superior Faculty Members." *Journal of Higher Education* 23 (7): 377–82.
- Young, P. 2006. "Out of Balance: Lecturers' Perceptions of Differential Status and Rewards in Relation to Teaching and Research." *Teaching in Higher Education* 11 (2): 191–202.